Emil Kim

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Education

Pusan National University — Busan, South Korea
M.Sc. in Artificial Intelligence, Expected Feb. 2027 — Mar 2025 – Present
Graduate School Scholarship

Pusan National University — Busan, South Korea
B.Sc. in Computer Science and Engineering — Sep 2020 – Jan 2025
Samsung Dream Scholarship • Outstanding Graduate (Top 10 in Class)

Research Interests

Biosignal Analysis (EEG, ECG) • Multimodal Representation • Medical Image Analysis.

Research Experience

IC&ML Lab, Pusan National University — Busan, South Korea
Research Assistant — Sep 2024 – Present

- Developed a multimodal deep learning framework for EEG-to-fMRI translation, improving reconstruction quality while reducing model parameters by 3.2× and peak GPU memory usage by 2.3×.
- Designed, trained, and benchmarked Transformer and state-space architectures for multichannel time-series analysis, improving balanced accuracy by 6.6% and reducing inference latency by 7.1% across three EEG datasets.
- Developed a compact CNN–Transformer model for EEG seizure detection, achieving 99.55% accuracy on the CHB-MIT dataset with 238K parameters.
- Developed a CNN-based classifier for Parkinson's disease detection using SPECT images, with saliency maps for model interpretability.

Teaching Experience

Pusan National University — Busan, South Korea
Teaching Assistant — Introduction to Unix Programming (CB2001102-060) — Mar 2026 – Jun 2026

- Designed programming assignments, led weekly practice sessions, and mentored undergraduate students.

Publications

Conference Proceedings

- E. Kim and J. K. Gahm, "MARBLE: Lightweight EEG-to-fMRI Translation via Mamba-Attention and ROI-Conditioned Decoding," *MICCAI*, Strasbourg, France, Sep 2026. **[Paper] [Code] [Demo]**
- E. Kim and J. K. Gahm, "CGM-EEG: Cross-Gated Mamba for Spatio-Temporal EEG Representation Learning," *IEEE ISBI* (Oral), London, UK, Apr 2026. **[Paper] [Code]**
- E. Kim, Y. Kenes, S. D. Tran, and J. K. Gahm, "Multi-Scale Attention Network for EEG-based Seizure Detection," *Korea Computer Congress (한국소프트웨어종합학술대회)*, Jeju, South Korea, Jul 2025. **[Paper]**

Journal Articles

- E. Kim and J. K. Gahm, "HSST-EEG: A Hybrid State-Space and Transformer Architecture for EEG Decoding," *IEEE Access*, vol. 14, pp. 79188–79196, 2026. **[Paper] [Code]**
- E. Kim, Y. Kenes, S. D. Tran, and J. K. Gahm, "Multi-Scale EEG Feature Extraction with Attention Mechanism for Seizure Detection," *Journal of KIISE*, vol. 53, no. 6, pp. 497–506, 2026. **[Paper]**

Projects

Ziyatron AI Almaty Kazakhstan
AI Researcher Jan 2025 - Nov 2025

- Curated an EEG dataset from 92 pediatric subjects in collaboration with 2nd Pediatric Hospital in Almaty.
- Integrated an EEG-based seizure detection model with GPT-4o to develop an AI-assisted clinical decision support platform.
- Presented the project at Digital Bridge 2025 in Astana, Kazakhstan.

Awards & Scholarships

1st Place, Three Minute Thesis (3MT), Pusan National University, South Korea Aug 2026

- Presented EEG–fMRI translation research in a three-minute presentation for a non-specialist audience.

Best Paper Award, Physical AI International Forum (한국인공지는학회), Seoul, South Korea Nov 2025

- Recognized for *HSST-EEG: A Hybrid State-Space and Transformer Architecture for EEG Decoding*.

Graduate School Scholarship, Pusan National University, Graduate School of AI Mar 2025 – Present

Samsung Dream Scholarship, Samsung Dream Scholarship Foundation Sep 2024 – Dec 2025

- Selected as one of two recipients per university for outstanding international students.

Additional Information

Programming Skills: Python, C, Matlab, scikit-learn, NumPy, Pandas
Language: Russian (Native), English (IELTS 8.0), Korean (TOPIK Level 5)